

#This code will be used to demonstrate theorem 6.19. We will demonstrate that mI^2

$$= \frac{-((z+w)^2 - y^2) \cdot ((w+z)^3 - y^3)^2 \cdot ((x^3 - z^3) \cdot w^2 + (x^3 - w^3) \cdot z^2)^2}{z \cdot w \cdot (w+z)^4 \cdot ((x^3 - w^3) \cdot (z^3 - y^3) + (x^3 - z^3) \cdot (w^3 - y^3))^2}$$
 have a unique
critical point along the γI curve; (Part 3 – the critical point is at a local minimum)
 restart;

#Define m1

$$m1 := \left(\left(a^2 b^2 + \frac{1}{3} \right)^2 (a+b)^4 (a^9 b^4 + (-3 b^5 + 2 b^3) a^8 + (3 b^6 - 4 b^4 + b^2) a^7 + (-5 b^7 - 6 b^5 + b^3 + \right. \\
+ \frac{b(b^4 - 4 b^2 + 1)(b^2 + 1)^3 a^7}{2} + \frac{(b^4 + 1)(b^8 + 8 b^4 + 1) a^6}{6} \\
- \frac{b(b-1)^2(b+1)^2(b^2 + 1)^3 a^5}{2} - \frac{b^2(b^8 + 2 b^6 - 2 b^4 + 2 b^2 + 1) a^4}{2} \\
\left. - \frac{b^3(b^2 + 1)^3 a^3}{3} + a^2 b^6 - \frac{b^6}{3} \right)^2 (a^2 + 1)(b^2 + 1)(b a + 1)^2 \Big) \\
m1 := \left(\left(a^2 b^2 + \frac{1}{3} \right)^2 (a+b)^4 (a^9 b^4 + (-3 b^5 + 2 b^3) a^8 + (3 b^6 - 4 b^4 + b^2) a^7 + (-5 b^7 - 6 b^5 + b^3 + 2 b) a^6 \right. \\
+ (3 b^8 + 2 b^6 - 20 b^4 - 4 b^2 + 1) a^5 + (-b^9 + 12 b^5 - 6 b^3 - 3 b) a^4 + (-2 b^8 - 3 b^6 + 2 b^4 + 3 b^2) a^3 \\
+ (-3 b^7 - 5 b^3) a^2 + (-2 b^6 + 3 b^4) a - b^5)^2 \Big) / \\
\left(\left((b^8 - b^6 + b^4) a^{10} - \frac{b^2(b^8 - 2 b^6 + 8 b^4 - 2 b^2 + 1) a^8}{2} \right. \right. \\
+ \frac{b(b^4 - 4 b^2 + 1)(b^2 + 1)^3 a^7}{2} + \frac{(b^4 + 1)(b^8 + 8 b^4 + 1) a^6}{6} \\
- \frac{b(b-1)^2(b+1)^2(b^2 + 1)^3 a^5}{2} - \frac{b^2(b^8 + 2 b^6 - 2 b^4 + 2 b^2 + 1) a^4}{2} \\
\left. \left. - \frac{b^3(b^2 + 1)^3 a^3}{3} + a^2 b^6 - \frac{b^6}{3} \right)^2 (a^2 + 1)(b^2 + 1)(b a + 1)^2 \right)$$

#Define r1

$$r1 := \left(a^{32} b^{16} + b^{16} + (-4 b^{18} + 8 b^{16} - 4 b^{14}) a^{30} + (6 b^{20} - 32 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}) a^{28} \right. \\
+ (72 b^{20} - 76 b^{18} + 8 b^{16} + 692 b^{14} + 72 b^{12} - 1024 b^{10}) a^{26} + (b^{32} + 8 b^{30} + 20 b^{28} \\
- 120 b^{26} - 3426 b^{24} - 7288 b^{22} - 125524 b^{20} - 222104 b^{18} - 279672 b^{16} - 317080 b^{14} \\
\left. - 157780 b^{12} - 32376 b^{10} + 1950 b^8 + 1672 b^6 + 20 b^4 + 8 b^2 + 1) a^{16} + (-1016 b^{26} \right.$$

$$\begin{aligned}
& - 8440 b^{24} - 21976 b^{22} - 25848 b^{20} - 37428 b^{18} - 31000 b^{16} - 8756 b^{14} + 264 b^{12} - 216 b^{10} \\
& + 8 b^8 + 8 b^6) a^{10} + (-265 b^{24} - 1864 b^{22} + 2192 b^{20} + 4232 b^{18} + 142 b^{16} - 1912 b^{14} \\
& + 144 b^{12} - 72 b^{10} - 9 b^8) a^8 + (-512 b^{22} - 696 b^{20} + 692 b^{18} + 1288 b^{16} - 76 b^{14} \\
& + 72 b^{12}) a^6 + (-250 b^{20} - 288 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}) a^4 + (-4 b^{18} + 8 b^{16} \\
& - 4 b^{14}) a^2 - 2560 \left(b^2 - \frac{1}{5} \right) b^7 (b^2 - 2) (b^2 + 1)^4 a^{23} + 2 b^4 (b^{24} + 36 b^{22} - 2636 b^{20} \\
& - 6620 b^{18} - 27212 b^{16} - 64184 b^{14} - 54798 b^{12} - 25400 b^{10} - 9036 b^8 - 1372 b^6 + 180 b^4 \\
& + 36 b^2 + 1) a^{12} - 4 b^2 (b^{28} + 12 b^{26} - 155 b^{24} + 1018 b^{22} + 4315 b^{20} + 34120 b^{18} + 65514 b^{16} \\
& + 57358 b^{14} + 41514 b^{12} + 19656 b^{10} + 4187 b^8 + 378 b^6 + 37 b^4 + 12 b^2 + 1) a^{14} - 4 b^2 (b^{28} \\
& + 12 b^{26} + 37 b^{24} - 518 b^{22} - 2085 b^{20} + 17992 b^{18} + 38058 b^{16} + 78414 b^{14} + 92394 b^{12} \\
& + 36360 b^{10} + 5211 b^8 - 2118 b^6 - 539 b^4 + 76 b^2 + 1) a^{18} + 2 b^4 (b^{24} + 36 b^{22} + 180 b^{20} \\
& - 2396 b^{18} - 25164 b^{16} - 42552 b^{14} - 90894 b^{12} - 84792 b^{10} - 12876 b^8 - 348 b^6 + 3252 b^4 \\
& + 420 b^2 - 127) a^{20} + 8 \left(b^{20} + b^{18} - 27 b^{16} - 159 b^{14} + \frac{2163}{2} b^{12} - 1827 b^{10} - \frac{1037}{2} b^8 \right. \\
& + 2369 b^6 - 699 b^4 + b^2 + 1 \left. \right) b^6 a^{22} - 9 \left(b^{16} + 8 b^{14} - 16 b^{12} - 72 b^{10} + \frac{3698}{9} b^8 - 328 b^6 \right. \\
& - 528 b^4 + \frac{4168}{9} b^2 + 1 \left. \right) b^8 a^{24} - 1024 b^{15} (b^2 + 1)^4 a^7 - 2560 \left(b^6 - \frac{8}{5} b^4 + \frac{6}{5} b^2 \right. \\
& - 1 \left. \right) b^{11} (b^2 + 1)^4 a^{11} - 1024 \left(b^2 - \frac{1}{2} \right) b^{13} (b^2 + 1)^5 a^9 + 512 b^7 (b^{10} - 19 b^8 + 42 b^6 \\
& - 9 b^4 - 10 b^2 + 1) (b^2 + 1)^4 a^{15} - 1536 \left(b^8 - b^6 + \frac{5}{3} b^4 - \frac{16}{3} b^2 + \frac{5}{3} \right) b^9 (b^2 + 1)^4 a^{13} \\
& + 512 b^7 (b^{10} - 22 b^8 + 81 b^6 - 78 b^4 + 11 b^2 + 1) (b^2 + 1)^4 a^{17} - 3584 b^7 \left(b^8 - \frac{52}{7} b^6 \right. \\
& + \frac{95}{7} b^4 - \frac{43}{7} b^2 + \frac{3}{7} \left. \right) (b^2 + 1)^4 a^{19} + 5632 \left(b^6 - \frac{42}{11} b^4 + \frac{38}{11} b^2 - \frac{5}{11} \right) b^7 (b^2 \\
& + 1)^4 a^{21} \Big):
\end{aligned}$$

#Calculating the derivative of $m1$ at $b=1$

$$d := \text{eval}(\text{diff}(m1, a), b=1) \cdot \left(-\frac{\text{eval}(\text{diff}(r1, b), b=1)}{\text{eval}(\text{diff}(r1, a), b=1)} \right) + \text{eval}(\text{diff}(m1, b), b=1) :$$

#Locate a range for $\mathcal{A}(1)$

$$\text{sturm}\left(\text{sturmseq}(\text{eval}(r1, b=1), a), a, \frac{2662}{1000}, \frac{2663}{1000}\right); \text{\#number of roots in } [2.662, 2.663)$$

1

(2)

#Define numerd

$$\text{numer}d := \text{numer}(d)$$

$$\begin{aligned} \text{numer}d := & -(3a^2 + 1)(a + 1)(a^9 - a^8 - 8a^6 - 18a^5 + 2a^4 - 8a^2 + a - 1)a(36a^{54} \\ & + 36a^{53} - 45a^{52} - 204a^{51} - 24a^{50} - 1200a^{49} - 6539a^{48} - 5072a^{47} + 10232a^{46} \\ & + 880584a^{45} + 997242a^{44} - 552632a^{43} - 16874480a^{42} - 8279792a^{41} - 25290786a^{40} \\ & + 27419184a^{39} - 185488852a^{38} + 323670044a^{37} + 103772737a^{36} + 330363436a^{35} \\ & + 269357112a^{34} - 1149358432a^{33} + 1052625423a^{32} - 2415483040a^{31} + 2354166736a^{30} \\ & - 2466496784a^{29} + 2108464828a^{28} - 1175180176a^{27} + 583692128a^{26} + 900592800a^{25} \\ & - 1362690316a^{24} + 2015306464a^{23} - 2133384772a^{22} + 2083790492a^{21} - 1699881691a^{20} \\ & + 1135358156a^{19} - 796335784a^{18} + 408694288a^{17} - 236890045a^{16} + 16756336a^{15} \\ & - 4151368a^{14} - 25201208a^{13} - 8331222a^{12} - 672056a^{11} - 3868016a^{10} - 1185200a^9 \\ & + 135246a^8 - 419600a^7 - 14220a^6 + 3236a^5 - 13721a^4 + 340a^3 + 136a^2 + 25) \end{aligned} \quad (3)$$

#Define denomd

$$\text{denom}d := \text{denom}(d)$$

$$\begin{aligned} \text{denom}d := & 2(3a^{10} - 9a^8 - 24a^7 + 10a^6 - 6a^4 - 8a^3 + 3a^2 - 1)^3(a^2 + 1)^2a^3(a^{28} - 7a^{24} \\ & - 208a^{22} + 405a^{20} + 23552a^{19} + 1232a^{18} + 10752a^{17} - 319075a^{16} - 48640a^{15} \\ & - 592416a^{14} - 26112a^{13} - 570845a^{12} + 23040a^{11} - 398944a^{10} + 19968a^9 - 143253a^8 \\ & + 5632a^7 - 42000a^6 - 4608a^5 + 647a^4 - 3584a^3 + 144a^2 - 65) \end{aligned} \quad (4)$$

#Verify that numerd and denomd have the same sign in the interval.

$$\text{sturm}\left(\text{sturmseq}(\text{numer}d, a), a, \frac{2662}{1000}, \frac{2663}{1000}\right); \text{\#number of roots in } [2.662, 2.663);$$

$$\text{sturm}\left(\text{sturmseq}(\text{denom}d, a), a, \frac{2662}{1000}, \frac{2663}{1000}\right); \text{\#number of roots in } [2.662, 2.663)$$

0

0

(5)

#Calculate the sign of $mI'(\gamma(1), 1)$ using $QI'(\frac{2662}{1000}, 1)$

$$\frac{\text{eval}\left(\text{numerd}, a = \frac{2662}{1000}\right)}{\text{eval}\left(\text{denomd}, a = \frac{2662}{1000}\right)}$$

— 16667448370201843655935712779480367894453229090712388291581874349888291394633312\ (6)
 06670126205228197411477772184775997590200823684800436051200485987259608727804\
 22918636862042660001084091232869190041665627/
 15156216351259885229540801081421334823768996035771963181632644319651051455252\
 41021640464311414017429433646234465005987471519121960084007973202048469461598\
 1892017492676548158168789960032074861463375000

restart;

$$\begin{aligned}
m1 := & \left(\left(a^2 b^2 + \frac{1}{3} \right)^2 (a+b)^4 (a^9 b^4 + (-3 b^5 + 2 b^3) a^8 + (3 b^6 - 4 b^4 + b^2) a^7 + (-5 b^7 - 6 b^5 \right. \\
& + b^3 + 2 b) a^6 + (3 b^8 + 2 b^6 - 20 b^4 - 4 b^2 + 1) a^5 + (-b^9 + 12 b^5 - 6 b^3 - 3 b) a^4 + (\\
& -2 b^8 - 3 b^6 + 2 b^4 + 3 b^2) a^3 + (-3 b^7 - 5 b^3) a^2 + (-2 b^6 + 3 b^4) a - b^5)^2 \Bigg) / \\
& \left(\left((b^8 - b^6 + b^4) a^{10} - \frac{b^2 (b^8 - 2 b^6 + 8 b^4 - 2 b^2 + 1) a^8}{2} \right. \right. \\
& + \frac{b (b^4 - 4 b^2 + 1) (b^2 + 1)^3 a^7}{2} + \frac{(b^4 + 1) (b^8 + 8 b^4 + 1) a^6}{6} \\
& - \frac{b (b-1)^2 (b+1)^2 (b^2 + 1)^3 a^5}{2} - \frac{b^2 (b^8 + 2 b^6 - 2 b^4 + 2 b^2 + 1) a^4}{2} \\
& \left. \left. - \frac{b^3 (b^2 + 1)^3 a^3}{3} + a^2 b^6 - \frac{b^6}{3} \right)^2 (a^2 + 1) (b^2 + 1) (b a + 1)^2 \right) \\
m1 := & \left(\left(a^2 b^2 + \frac{1}{3} \right)^2 (a+b)^4 (a^9 b^4 + (-3 b^5 + 2 b^3) a^8 + (3 b^6 - 4 b^4 + b^2) a^7 + (-5 b^7 - 6 b^5 \right. \textbf{(7)} \\
& + b^3 + 2 b) a^6 + (3 b^8 + 2 b^6 - 20 b^4 - 4 b^2 + 1) a^5 + (-b^9 + 12 b^5 - 6 b^3 - 3 b) a^4 + (\\
& -2 b^8 - 3 b^6 + 2 b^4 + 3 b^2) a^3 + (-3 b^7 - 5 b^3) a^2 + (-2 b^6 + 3 b^4) a - b^5)^2 \Bigg) / \\
& \left(\left((b^8 - b^6 + b^4) a^{10} - \frac{b^2 (b^8 - 2 b^6 + 8 b^4 - 2 b^2 + 1) a^8}{2} \right. \right. \\
& + \frac{b (b^4 - 4 b^2 + 1) (b^2 + 1)^3 a^7}{2} + \frac{(b^4 + 1) (b^8 + 8 b^4 + 1) a^6}{6} \\
& - \frac{b (b-1)^2 (b+1)^2 (b^2 + 1)^3 a^5}{2} - \frac{b^2 (b^8 + 2 b^6 - 2 b^4 + 2 b^2 + 1) a^4}{2} \\
& \left. \left. - \frac{b^3 (b^2 + 1)^3 a^3}{3} + a^2 b^6 - \frac{b^6}{3} \right)^2 (a^2 + 1) (b^2 + 1) (b a + 1)^2 \right)
\end{aligned}$$

#Define r1

$$\begin{aligned}
r1 := & \left(a^{32} b^{16} + b^{16} + (-4 b^{18} + 8 b^{16} - 4 b^{14}) a^{30} + (6 b^{20} - 32 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}) a^{28} \right. \\
& + (72 b^{20} - 76 b^{18} + 8 b^{16} + 692 b^{14} + 72 b^{12} - 1024 b^{10}) a^{26} + (b^{32} + 8 b^{30} + 20 b^{28} \\
& - 120 b^{26} - 3426 b^{24} - 7288 b^{22} - 125524 b^{20} - 222104 b^{18} - 279672 b^{16} - 317080 b^{14} \\
& \left. - 157780 b^{12} - 32376 b^{10} + 1950 b^8 + 1672 b^6 + 20 b^4 + 8 b^2 + 1) a^{16} + (-1016 b^{26} \right.
\end{aligned}$$

$$\begin{aligned}
& - 8440 b^{24} - 21976 b^{22} - 25848 b^{20} - 37428 b^{18} - 31000 b^{16} - 8756 b^{14} + 264 b^{12} - 216 b^{10} \\
& + 8 b^8 + 8 b^6) a^{10} + (-265 b^{24} - 1864 b^{22} + 2192 b^{20} + 4232 b^{18} + 142 b^{16} - 1912 b^{14} \\
& + 144 b^{12} - 72 b^{10} - 9 b^8) a^8 + (-512 b^{22} - 696 b^{20} + 692 b^{18} + 1288 b^{16} - 76 b^{14} \\
& + 72 b^{12}) a^6 + (-250 b^{20} - 288 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}) a^4 + (-4 b^{18} + 8 b^{16} \\
& - 4 b^{14}) a^2 - 2560 \left(b^2 - \frac{1}{5} \right) b^7 (b^2 - 2) (b^2 + 1)^4 a^{23} + 2 b^4 (b^{24} + 36 b^{22} - 2636 b^{20} \\
& - 6620 b^{18} - 27212 b^{16} - 64184 b^{14} - 54798 b^{12} - 25400 b^{10} - 9036 b^8 - 1372 b^6 + 180 b^4 \\
& + 36 b^2 + 1) a^{12} - 4 b^2 (b^{28} + 12 b^{26} - 155 b^{24} + 1018 b^{22} + 4315 b^{20} + 34120 b^{18} + 65514 b^{16} \\
& + 57358 b^{14} + 41514 b^{12} + 19656 b^{10} + 4187 b^8 + 378 b^6 + 37 b^4 + 12 b^2 + 1) a^{14} - 4 b^2 (b^{28} \\
& + 12 b^{26} + 37 b^{24} - 518 b^{22} - 2085 b^{20} + 17992 b^{18} + 38058 b^{16} + 78414 b^{14} + 92394 b^{12} \\
& + 36360 b^{10} + 5211 b^8 - 2118 b^6 - 539 b^4 + 76 b^2 + 1) a^{18} + 2 b^4 (b^{24} + 36 b^{22} + 180 b^{20} \\
& - 2396 b^{18} - 25164 b^{16} - 42552 b^{14} - 90894 b^{12} - 84792 b^{10} - 12876 b^8 - 348 b^6 + 3252 b^4 \\
& + 420 b^2 - 127) a^{20} + 8 \left(b^{20} + b^{18} - 27 b^{16} - 159 b^{14} + \frac{2163}{2} b^{12} - 1827 b^{10} - \frac{1037}{2} b^8 \right. \\
& + 2369 b^6 - 699 b^4 + b^2 + 1 \left. \right) b^6 a^{22} - 9 \left(b^{16} + 8 b^{14} - 16 b^{12} - 72 b^{10} + \frac{3698}{9} b^8 - 328 b^6 \right. \\
& - 528 b^4 + \frac{4168}{9} b^2 + 1 \left. \right) b^8 a^{24} - 1024 b^{15} (b^2 + 1)^4 a^7 - 2560 \left(b^6 - \frac{8}{5} b^4 + \frac{6}{5} b^2 \right. \\
& - 1 \left. \right) b^{11} (b^2 + 1)^4 a^{11} - 1024 \left(b^2 - \frac{1}{2} \right) b^{13} (b^2 + 1)^5 a^9 + 512 b^7 (b^{10} - 19 b^8 + 42 b^6 \\
& - 9 b^4 - 10 b^2 + 1) (b^2 + 1)^4 a^{15} - 1536 \left(b^8 - b^6 + \frac{5}{3} b^4 - \frac{16}{3} b^2 + \frac{5}{3} \right) b^9 (b^2 + 1)^4 a^{13} \\
& + 512 b^7 (b^{10} - 22 b^8 + 81 b^6 - 78 b^4 + 11 b^2 + 1) (b^2 + 1)^4 a^{17} - 3584 b^7 \left(b^8 - \frac{52}{7} b^6 \right. \\
& + \frac{95}{7} b^4 - \frac{43}{7} b^2 + \frac{3}{7} \left. \right) (b^2 + 1)^4 a^{19} + 5632 \left(b^6 - \frac{42}{11} b^4 + \frac{38}{11} b^2 - \frac{5}{11} \right) b^7 (b^2 \\
& + 1)^4 a^{21} \Big):
\end{aligned}$$

#Calculating the derivative of m1 at b=3.7

$$d := eval\left(diff(m1, a), b = \frac{37}{10}\right) \cdot \left(- \frac{eval\left(diff(r1, b), b = \frac{37}{10}\right)}{eval\left(diff(r1, a), b = \frac{37}{10}\right)} \right) + eval\left(diff(m1, b), b = \frac{37}{10}\right) :$$

$$\begin{aligned} & \# \text{Locate a range for } \gamma l \left(\frac{37}{10} \right) \\ & \text{sturm} \left(\text{sturmseq} \left(\text{eval} \left(r1, b = \frac{37}{10} \right), a \right), a, \frac{37}{10}, \frac{38}{10} \right); \# \text{number of roots in } [3.7, 3.8) \\ & \quad \quad \quad 1 \end{aligned} \tag{8}$$

$$\begin{aligned} & \# \text{Define numerd} \\ & \text{numer}d := \text{numer}(d) : \\ & \# \text{Define denomd} \\ & \text{denom}d := \text{denom}(d) : \\ & \# \text{Verify that numerd and denomd have the same sign in the interval.} \\ & \text{sturm} \left(\text{sturmseq}(\text{numer}d, a), a, \frac{37}{10}, \frac{38}{10} \right); \# \text{number of roots in } [3.7, 3.8); \\ & \text{sturm} \left(\text{sturmseq}(\text{denom}d, a), a, \frac{37}{10}, \frac{38}{10} \right); \# \text{number of roots in } [3.7, 3.8) \\ & \quad \quad \quad 0 \\ & \quad \quad \quad 0 \end{aligned} \tag{9}$$

$$\begin{aligned} & \# \text{Calculate the sign of } m l' \left(\gamma l \left(\frac{38}{10} \right), \frac{38}{10} \right) \text{ using } m l' \left(\frac{38}{10}, \frac{38}{10} \right) \\ & \frac{\text{eval} \left(\text{numer}d, a = \frac{38}{10} \right)}{\text{eval} \left(\text{denom}d, a = \frac{38}{10} \right)} \\ & 1112308828725666873502629642887511319433139335883748932268020542966651415612050 \backslash \tag{10} \\ & 3075321509878502242615981051797296841144355261588718738055977861414507507031 \backslash \\ & 25000000 / \\ & 7473635688809379162908481016145450327406785222804342395051225057071356710324 \backslash \\ & 8724552743385028720026018130199445234110023410473002226877674669169710793939 \backslash \\ & 1108942853343 \end{aligned}$$